

Procurement Bill of Materials

Module-level ordering guide (no vendor links)

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Platform: esp32

Total line items: 17

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Procurement note: Verify connector orientation, voltage levels, and form factor before ordering.

#	Item	Qty	Exactness	Ref	Required specs
1	ESP32 DevKit v1	1	Compatible	esp32	This project targets the ESP32 DevKit v1.
2	DFRobot SEN0142 Fermion MPU-6050 6 DOF Sensor Breakout	1	Exact	SEN0142	DFRobot SEN0142 MPU-6050 breakout with 3-5 V board input, I2C interface, onboard I2C pull-ups, and i2cdevlib Arduino example coverage.
3	u-blox NEO-M8N GPS Module	1	Exact	ublox-neo-m8n-gps- module	u-blox NEO-M8N GNSS receiver module. Outputs NMEA data over UART by default and may expose PPS on some breakouts. Use 3.3V logic for MCU UART pins.
4	Push Button	1	Exact	DFR0029-W	Momentary push button switch
5	Push Button	1	Exact	DFR0029-W	Momentary push button switch
6	Push Button	1	Exact	DFR0029-W	Momentary push button switch
7	Buzzer	1	Exact	1536	Piezo buzzer for sound output
8	MicroSD Card Module	1	Exact	0700598870473	SPI-based microSD card adapter module for SPI-capable microcontrollers. Uses MOSI, MISO, SCK, and CS plus power and ground. Many low-cost modules include a 3.3 V regulator and level shifting for 5 V MCU boards, while bare breakouts should be powered and signalled at 3.3 V.
9	Waveshare SIM7600G-H 4G LTE/GNSS development module	1	Generic	—	A preassembled 4G cellular modem board that sends SMS messages and has an onboard wide-input power supply and 3.3 V compatible UART.
10	12 V amber LED beacon module with logic-level MOSFET driver	1	Generic	—	A weather-resistant amber emergency light driven through a protected low-side MOSFET module.
11	PC817 isolated headlight-assist relay module	1	Generic	—	An opto-isolated relay contact that parallels the tricycle headlight switch without electrically joining vehicle lamp wiring to the ESP32.
12	I2C RGB status indicator module	1	Generic	—	A small three-colour status light module that shares the I2C sensor wires and shows normal, warning, fault, and power states.
13	12 V vehicle input protection and 5 V 5 A buck converter module	1	Generic	—	A fused vehicle-input power module with reverse-polarity protection, TVS surge clamp, and a regulated 5 V output for the portable unit.
14	Protected 1-cell Li-ion UPS/charger with 5 V 3 A boost output	1	Generic	—	A protected 3.7 V backup-battery charger and automatic load-sharing boost module that keeps the alert device powered after vehicle input is lost.
15	Protected 3.7 V 3000 mAh 18650 Li-ion battery pack	1	Generic	—	A protected rechargeable backup battery that powers the device briefly during loss of vehicle power.

#	Item	Qty	Exactness	Ref	Required specs
16	100 k Ω and 27 battery-voltage divider module	1 Ω	Generic	—	A preassembled resistor divider that safely scales the single-cell battery voltage for an ESP32 analogue input.
17	100 k Ω and 22 vehicle-voltage divider with 3.3 V clamp	1 Ω	Generic	—	A protected resistor divider that lets the controller sense whether the fused tricycle supply is present without exposing an ESP32 pin to 12 V.